# EEPC2202 Electrical Machine-I Laboratory (0-0-3)

| Sl. No | Name of the Experiment | Hrs. |
|---|---|---|
| **1.** | Load characteristics of DC shunt/compound generator. | **3** |
| **2.** | Load characteristics of DC series motor. | **3** |
| **3.** | Speed control of DC shunt motor by armature voltage control and flux control method. | **3** |
| **4.** | Determination of critical resistance and critical speed from no load test of a DC shunt generator. | **3** |
| **5.** | Back-to-Back testing of DC machines (Hopkinson's method) | **3** |
| **6.** | Determination of efficiency and voltage regulation by open circuit and short circuit test on single-phase transformer. | **3** |
| **7.** | Parallel operation of two single phase transformers. | **3** |
| **8.** | Study of Open delta and Scott connection of two single phase transformers. | **3** |
| **9.** | To connect and measure the line voltage & phase voltage of three single-phase transformer in<br>(a) star-star (Y-Y)<br>(b) star-delta (Y- $\Delta$)<br>(c) delta-delta ($\Delta$-$\Delta$)<br>(d) delta-star ($\Delta$-Y) | **3** |
| **10.** | Back-to Back test on two single phase transformers. | **3** |

**CO1:** Understand load characteristics of DC shunt and compound generators and series motors.
**CO2:** Learn speed control methods for DC shunt motors, including armature voltage and flux control.
**CO3:** Analyse critical resistance and critical speed determination from DC shunt generator no-loadtests.
**CO4:** Explore back-to-back testing of DC machines using Hopkinson's method.
**CO5:** Determine efficiency and voltage regulation through open circuit and short circuit tests on single-phase transformers, and study parallel operation and different connection configurations of single-phase transformers.